Terra Nova Learning Academy – Executive Report

Prepared for**: Terra Nova Learning Academy (TNLA)**
Prepared by: **Team M4 (Muhammad Hadi Hussain, Kripi Joshi, Carlos Zuniga, Yufei Ye)**
Date**: December 7, 2025**

ABSTRACT

Terra Nova Learning Academy (TNLA) aims to strengthen workforce capability by ensuring that its training investments lead to meaningful engagement and skill development. Using a motivation survey linked to training outcomes, we identified three underlying motivation themes and used them to segment employees into three distinct learner groups. While all segments show different levels of engagement and proficiency growth, one pattern stands out: Segment C—"Structure-Seeking Traditionalists"—shows the highest potential for improvement but is highly sensitive to course modality. They struggle in asynchronous environments yet demonstrate strong capability gains when guided through structured, instructor-led learning.

Aligning training design, communication, and learner support around these insights will allow TNLA to improve completion rates, enhance learner experience, and maximize return on training investments.

1. Business Context and Objectives

TNLA operates in a dynamic learning environment with employees who vary widely in their motivations, expectations, and preferred learning formats. To strengthen the impact of its training ecosystem, TNLA needs a deeper understanding of what drives employees to engage, how different motivational patterns translate into behavior, and which training formats best support learner success.

Purpose of This Analysis

To address this, TNLA surveyed employees using a 12-item motivation questionnaire and linked responses to training completion, modality choices, and learning gains. This report answers four key questions:

1. **What motivates TNLA employees to participate in training?**
2. **How do these motivations cluster into meaningful learner profiles?**
3. **How do learner segments differ in outcomes such as completion and learning gains?**
4. **How can TNLA adapt its training strategy to better support its most impactful learner segments?**

Ultimately, this work enables TNLA to align its learning strategy with employee needs, ensuring training investments are efficient, high-impact, and learner-centered.

2. Motivation Themes Identified in Survey Responses

The analysis of Q1–Q12 revealed **three strong, conceptually meaningful motivation themes**, each capturing a unique dimension of how employees think about training.

Theme 1: Self-Directed & Format-Sensitive Learning

This theme reflects employees who approach learning proactively and independently. They demonstrate initiative in navigating their learning journey and are highly responsive to course modality.

Key Characteristics:

- Self-motivated and autonomous learners
- Comfortable exploring supplemental materials
- Sensitive to the design, pacing, and modality of a course
- Prefer the flexibility of digital and self-paced formats

Interpretation:
For these learners, autonomy drives engagement. Their training success depends on having access to well-designed, flexible, and modular learning experiences that allow them to control pace and path.

Theme 2: Feedback & Support Environment

This theme highlights learners who depend on social reinforcement and structured feedback loops.

Key Characteristics:

- Seek clarity in expectations and skill application
- Value manager reinforcement and instructor acknowledgment
- Benefit from opportunities for discussion, coaching, and reflection

Interpretation:
These learners thrive when they feel supported. They need visibility into how learning connects to work and appreciate feedback mechanisms that validate their progress.

Theme 3: Confidence & Career-Building Orientation

This theme captures learners who see training as a career accelerator.

Key Characteristics:

- Desire to build confidence in core skills
- Motivated by credentials, skill advancement, and career progression
- Look for training tied to long-term professional growth

Interpretation:
They are motivated when training has a visible impact. These learners respond strongly to programs that offer clear value propositions, badges, or certifications signaling advancement.

3. Employee Segments

Using these motivation themes as a basis, three distinct segments of TNLA learners emerged. These segments highlight meaningful differences in learning behavior, expectations, and outcomes.

Segment A: Struggling but Committed Learners (80 learners)

Profile: Motivated to learn but require structure, clarity, and pacing support to stay on track.
Training Behavior: Moderate completion (78.8%) but negative learning gains (–2.542), especially in less structured environments.
Why They Matter: A large segment whose performance can improve significantly with clearer scaffolding and guided course design.

Segment B: Self-Directed High Performers (61 learners)

Profile: Independent, confident learners who excel with flexibility and minimal guidance.
Training Behavior: Highest completion (93.4%) and strong learning gains (+7.116) across all modalities.
Why They Matter: Low-touch, reliable performers who enable scalable digital learning with minimal instructional resources.

Segment C: Structure-Seeking Traditionalists (59 learners)

Profile: Prefer clear sequencing, instructor presence, and predictable workflows.
Training Behavior: Underperform in unstructured or asynchronous formats but show some of the highest learning gains when placed in structured, instructor-led environments.
Why They Matter: This segment has the highest potential for improvement due to its strong sensitivity to instructional structure.

4. Differences in Training Outcomes Across Segments

Segment-level differences highlight how motivation patterns translate into engagement and performance.

Completion Rates Differences

Descriptive completion averages show clear separation:

- **Segment B (Self-Directed High Performers): 93.4%** completion
- **Segment A (Struggling but Committed): 78.8%**
- **Segment C (Structure-Seeking Traditionalists): 79.7%**

Segment B is consistently engaged regardless of format. Segments A and C complete training at materially lower rates, suggesting that environmental design plays a larger role in their persistence.

Learning Gains

Average learning gains reveal meaningful gaps in capability building:

- **Segment B: +7.116** (strongest learning outcomes)
- **Segment A: –2.542**
- **Segment C: –3.219**
- **Segment B** again shows the strongest average learning gains, performing well regardless of modality.
- **Segments A and C** show negative or modest gains overall, but Segment C demonstrates dramatic improvement when placed in instructor-led formats.
- Regression results show Segment C's predicted learning gains shift dramatically by modality: –17.9 in asynchronous courses and +7.71 in instructor-led formats, confirming that instructional structure is the key driver of their performance.

Modality Impact

Enrollment patterns reveal mismatch between learner needs and training design:

Segment	Async	In-Person	Online-Live
Segment A	21.2%	36.2%	42.5%
Segment B	23.0%	39.3%	37.7%
Segment C	23.7%	33.9%	42.4%

Critically, Segment C disproportionately enrolls in **online-live and async** formats.

4.4 Implications

Across all approaches, the same pattern emerges:

- **Segment B succeeds regardless of format.**
- **Segment A struggles in digital settings but can stabilize with moderate structure.**
- **Segment C shows the most dramatic improvement when providing predictable, guided, instructor-led environments.**

This creates a clear opportunity for TNLA to re-align training design to where it creates the greatest uplift.

Segment C is prioritized because it shows the largest performance gap between asynchronous (–17.9 predicted) and structured instructor-led (+7.71 predicted) modalities.

5. Recommendations for Target Segment: Segment C – Structure-Seeking Traditionalists

Segment C shows the strongest dependence on structured, instructor-led learning environments. Their performance varies dramatically by modality, with analysis results indicating **large negative learning gains in asynchronous formats (–17.9 predicted)** and **significantly positive gains in instructor-led formats (+7.71 predicted)**. To unlock this segment's potential, TNLA should redesign both delivery and communication strategies to better match their learning needs.

(a) Course and Modality Guidance

Encourage: Instructor-Led & Structured Blended Courses

TNLA should guide Segment C toward:

- **In-person instructor-led courses**, which provide the highest predicted uplift in learning (+7.71)
- **Structured blended formats**, with fixed pacing, defined weekly modules, and facilitator checkpoints
- **Courses with clearly sequenced agendas and predictable workflows**

(b) Communication and Positioning

TNLA should frame courses for Segment C by emphasizing:

- Step-by-step progression
- Predictable agendas and clear timelines
- Regular instructor interaction
- How structured learning directly improves confidence and mastery

This reduces uncertainty and encourages enrollment in the formats where they succeed.

(c) Engagement and Support Strategies

To improve success for Segment C, TNLA should implement:

1. **Guided Learning Pathways**
   – Weekly checkpoints, pacing guides, and visible progress indicators.
2. **Instructor Touchpoints**
   – Short live Q&A sessions, office hours, and initiative-taking reminders.
3. **Peer or Cohort Accountability**
   – Small learning groups or discussion forums to reinforce structure.
4. **Embedded Scaffolding**
   – Worked examples, rubrics, and quick feedback mechanisms.

These supports create structure even within semi-flexible formats.

(d) Risks and Considerations

- **Resource intensity:** Instructor-led formats require more staffing and scheduling.
- **Scalability:** Structured models may not scale quickly across all regions.
- **Behavioral resistance:** Some employees may prefer flexibility.
- **Segmentation stability:** Learner preferences may evolve; segmentation should be refreshed periodically.

A phased rollout—focused first on high-volume or mission-critical courses—can mitigate these risks.

6. Closing Summary

The analysis demonstrates that aligning training design with learner motivations and modality preferences can materially improve TNLA's workforce capability.
By tailoring structured, instructor-led experiences for Segment C, TNLA's most modality-sensitive group, the organization can significantly increase completion rates and learning gains while improving the overall consistency of capability building.

These insights provide a clear roadmap for high-impact, data-driven improvements to TNLA's learning ecosystem.

Appendices

Appendix A – PCA Outputs

• PCA Factor Loadings on Survey Questions

```
# ---- 5. PCA loadings ----

loadings = pd.DataFrame(
    pca.components_.T,
    index=motivation_cols,
    columns=[f"PC{i}" for i in range(1, len(motivation_cols)+1)
)

loadings.iloc[:, :5].round(3)
```

✓ 0.0s

	PC1	PC2	PC3	PC4	PC5
Q1	0.086	0.638	0.106	-0.254	-0.051
Q2	0.358	0.095	0.210	0.214	0.034
Q3	0.259	-0.082	0.559	0.245	-0.072
Q4	0.341	0.000	-0.236	-0.322	-0.130
Q5	-0.164	0.238	-0.359	0.725	-0.395
Q6	0.363	0.060	-0.241	0.048	-0.135
Q7	0.254	-0.032	-0.539	-0.036	0.163
Q8	-0.000	0.671	0.133	-0.086	-0.019
Q9	-0.320	0.084	-0.207	-0.202	0.312
Q10	0.388	0.063	-0.176	-0.067	-0.158
Q11	0.250	0.133	-0.039	0.371	0.799
Q12	0.378	-0.193	0.075	-0.078	-0.110

• Explained Variance Ratios

- PC1: 44.14%
- PC2: 15.69%
- PC3: 9.96%
- Cumulative (PC1–PC3): 69.78%

• Interpretation: Motivational themes representing self-directed growth, support needs, and efficiency orientation.

	PC1	PC2	PC3
0	-2.032470	1.153019	-1.707330
1	1.234600	-0.398897	0.474666
2	3.388966	-0.702441	0.611477
3	-0.409249	-1.155489	-0.202061
4	-2.036008	1.195319	1.273887

```
explained = pca.explained_variance_ratio_
print("Explained variance ratio:", explained)
print("Cumulative:", explained.cumsum())
```

✓ 0.0s

```
Explained variance ratio: [0.44135362 0.15685331 0.09957838 0.06913754 0.05973918 0.04081721
 0.04003325 0.02957979 0.0230253  0.01645078 0.0121737  0.01125794]
Cumulative: [0.44135362 0.59820693 0.69778531 0.76692284 0.82666202 0.86747923
 0.90751248 0.93709227 0.96011757 0.97656836 0.98874206 1.        ]
```

Appendix B – Cluster Summary Table

Segment | Size | Completion Rate | Avg Learning Gain | Key Insight

Segment A | 80 | 0.788 | –2.542 | Needs structure; struggles in online-live
Segment B | 61 | 0.934 | +7.116 | High performers across all modalities
Segment C | 59 | 0.797 | –3.219 | Highly modality-sensitive; large gains in guided formats

```
# ---- 9. Cluster summaries: motivations & outcomes ----

summary_motivation = full.groupby("Cluster3")[motivation_cols].mean().round(2)
summary_outcomes = full.groupby("Cluster3")[["CompletionFlag",
                                             "Intake_Proficiency",
                                             "Outcome_Proficiency",
                                             "LearningGain"]].mean().round(2)
counts = full["Cluster3"].value_counts().rename("Count").sort_index()

print("Cluster sizes:")
display(counts)

print("\nAverage motivation item scores by cluster:")
display(summary_motivation)

print("\nTraining outcomes by cluster:")
display(summary_outcomes)
```

0.0s

```
luster sizes:

luster3
    80
    61
    59
ame: Count, dtype: int64


verage motivation item scores by cluster:
```

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
Cluster3												
0	3.62	2.39	2.56	1.91	3.30	2.21	2.58	3.88	3.51	1.84	2.70	1.11
1	3.72	4.21	3.43	3.52	2.82	3.93	3.36	3.44	2.51	4.07	3.38	3.98
2	1.81	2.76	2.86	2.20	3.02	2.63	2.88	2.31	3.12	2.29	2.75	2.31

```
raining outcomes by cluster:
```

	CompletionFlag	Intake_Proficiency	Outcome_Proficiency	LearningGain
Cluster3				
0	0.79	59.95	57.40	-2.54
1	0.93	60.63	67.74	7.12
2	0.80	60.58	57.36	-3.22

Appendix C – Modality Sensitivity (Regression Results)
Predicted Learning Gains (OLS Interaction Model):

- Async: –17.9 for Segment C
- Instructor-led: +7.71 for Segment C
- Segment B consistently positive across modalities
- Segment A negative in online-live formats

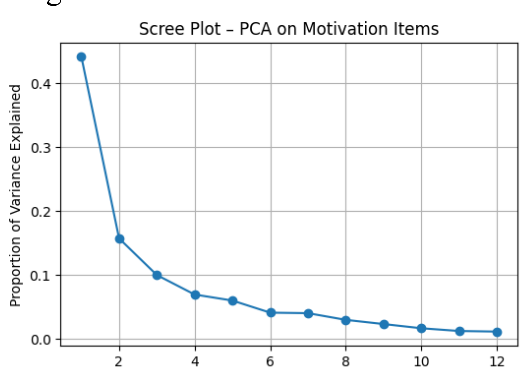
```
                              OLS Regression Results
==============================================================================
Dep. Variable:           LearningGain   R-squared:                       0.079
Model:                            OLS   Adj. R-squared:                  0.040
Method:                 Least Squares   F-statistic:                     2.040
Date:                Sat, 06 Dec 2025   Prob (F-statistic):             0.0438
Time:                        23:14:38   Log-Likelihood:                -939.90
No. Observations:                 200   AIC:                             1898.
Df Residuals:                     191   BIC:                             1927.
Df Model:                           8
Covariance Type:            nonrobust
========================================================================================================================
                                                      coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------------------------------
Intercept                                          -0.6235      6.599     -0.094      0.925     -13.641      12.394
C(Cluster3)[T.1]                                    6.7592      9.820      0.688      0.492     -12.611      26.129
C(Cluster3)[T.2]                                  -17.2836      9.820     -1.760      0.080     -36.654       2.086
C(CourseModality)[T.In-Person]                      3.7822      8.312      0.455      0.650     -12.612      20.176
C(CourseModality)[T.Online Live]                   -7.7412      8.083     -0.958      0.339     -23.684       8.201
C(Cluster3)[T.1]:C(CourseModality)[T.In-Person]    -0.1304     12.362     -0.011      0.992     -24.514      24.253
C(Cluster3)[T.2]:C(CourseModality)[T.In-Person]    21.8400     12.609      1.732      0.085      -3.031      46.711
C(Cluster3)[T.1]:C(CourseModality)[T.Online Live]   6.5315     12.264      0.533      0.595     -17.659      30.722
C(Cluster3)[T.2]:C(CourseModality)[T.Online Live]  21.9083     12.158      1.802      0.073      -2.074      45.890
==============================================================================
Omnibus:                       63.244   Durbin-Watson:                   1.918
```

Predicted Completion Probabilities (Logit Model):
• Segment B remains high across modalities
• Segment A decreases in online-live
• Segment C increases sharply with more structure

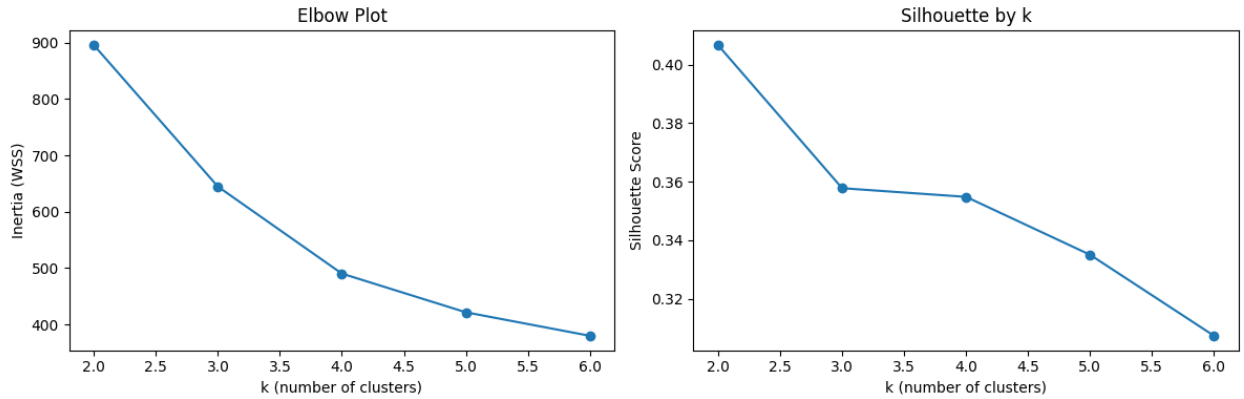
```
Optimization terminated successfully.
         Current function value: 0.403746
         Iterations 7
                           Logit Regression Results
==============================================================================
Dep. Variable:         CompletionFlag   No. Observations:                  200
Model:                          Logit   Df Residuals:                      191
Method:                           MLE   Df Model:                            8
Date:                Sat, 06 Dec 2025   Pseudo R-squ.:                 0.09852
Time:                        23:14:38   Log-Likelihood:                -80.749
converged:                       True   LL-Null:                       -89.574
Covariance Type:            nonrobust   LLR p-value:                   0.02402
========================================================================================================================
                                                      coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------------------------------------
Intercept                                           1.5404      0.636      2.421      0.015       0.293       2.787
C(Cluster3)[T.1]                                    1.0245      1.217      0.842      0.400      -1.361       3.410
C(Cluster3)[T.2]                                   -1.2528      0.835     -1.501      0.133      -2.888       0.383
C(CourseModality)[T.In-Person]                      0.2921      0.834      0.350      0.726      -1.342       1.926
C(CourseModality)[T.Online Live]                   -0.6650      0.739     -0.900      0.368      -2.114       0.784
C(Cluster3)[T.1]:C(CourseModality)[T.In-Person]     0.2784      1.678      0.166      0.868      -3.010       3.567
C(Cluster3)[T.2]:C(CourseModality)[T.In-Person]     2.3646      1.428      1.656      0.098      -0.434       5.163
C(Cluster3)[T.1]:C(CourseModality)[T.Online Live]   0.4514      1.473      0.306      0.759      -2.436       3.339
C(Cluster3)[T.2]:C(CourseModality)[T.Online Live]   1.7636      1.043      1.691      0.091      -0.281       3.808
```

Appendix D – Visual Analytics

• Figure A1: PCA Scree Plot



• Figure A2: K-means Clustering Elbow or Silhouette Method



- **Appendix E –Use of Gen AI**
  • Gen AI was primarily used in brainstorming ideas on how to first understand the case, its requirements and how the report needed to be written so that it is non-technical in nature
- Some python code for analysis was also developed with the help of Gen AI